

PART A

Circuit (a): Series Circuit

1. Total Resistance

$$R_{\text{total}} = 20\Omega + 30\Omega + 50\Omega = 100\Omega$$

2. Total Current

$$I = V / R$$

$$I = 125\text{V} / 100\Omega = 1.25 \text{ A}$$

3. Current through Each Resistor

$$I_1 = I_2 = I_3 = 1.25 \text{ A}$$

4. Voltage Drop Across Each Resistor

$$V_1 (20\Omega) = 1.25 \times 20 = 25 \text{ V}$$

$$V_2 (30\Omega) = 1.25 \times 30 = 37.5 \text{ V}$$

$$V_3 (50\Omega) = 1.25 \times 50 = 62.5 \text{ V}$$

Check:

$$25 + 37.5 + 62.5 = 125 \text{ V}$$

5. Power Dissipated

$$P_1 (20\Omega) = 25 \times 1.25 = 31.25 \text{ W}$$

$$P_2 (30\Omega) = 37.5 \times 1.25 = 46.875 \text{ W}$$

$$P_3 (50\Omega) = 62.5 \times 1.25 = 78.125 \text{ W}$$

Circuit (b): Parallel Circuit

1. Total Resistance

$$1 / R_{\text{total}} = (1 / 20) + (1 / 100) + (1 / 50)$$

$$1 / R_{\text{total}} = 0.05 + 0.01 + 0.02 = 0.08$$

$$R_{\text{total}} = 1 / 0.08 = 12.5\Omega$$

2. Total Current

$$I_{\text{total}} = V / R_{\text{total}}$$

$$I_{\text{total}} = 125\text{V} / 12.5\Omega = 10 \text{ A}$$

3. Current through Each Resistor

$$I_1 (20\Omega) = 125 / 20 = 6.25 \text{ A}$$

$$I_2 (100\Omega) = 125 / 100 = 1.25 \text{ A}$$

$$I_3 (50\Omega) = 125 / 50 = 2.5 \text{ A}$$

Check:

$$6.25 + 1.25 + 2.5 = 10 \text{ A}$$

4. Voltage Across Each Resistor

$$V_1 = V_2 = V_3 = 125 \text{ V}$$

5. Power Dissipated

$$P_1 (20\Omega) = 125 \times 6.25 = 781.25 \text{ W}$$

$$P_2 (100\Omega) = 125 \times 1.25 = 156.25 \text{ W}$$

$$P_3 (50\Omega) = 125 \times 2.5 = 312.5 \text{ W}$$